



Georgia Tech  
College of  
Engineering

# COE 3001

# MECHANICS OF DEFORMABLE BODIES

*Lecture 8* – Torsion – torsion formula

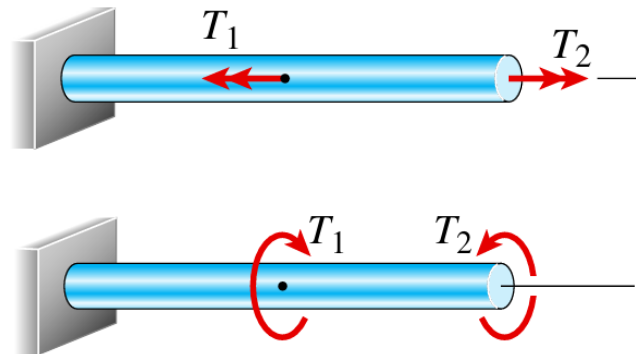
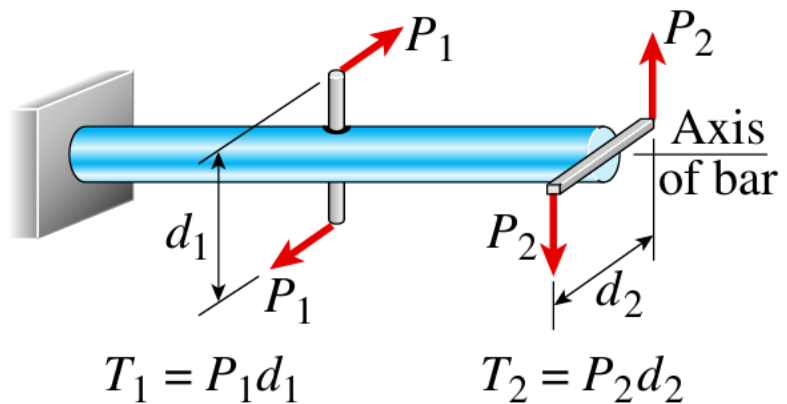
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Georgia Institute of Technology

Feb. 11, 2026

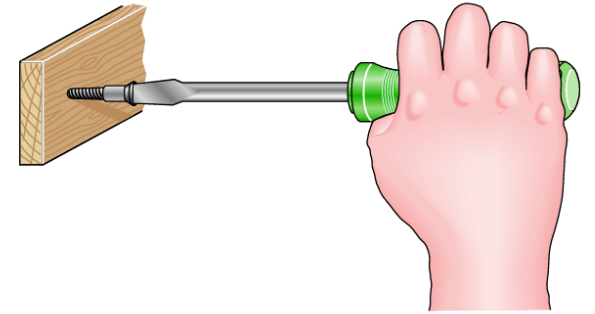
# Torsion

Torsion refers to the **twisting** of a straight bar when it is loaded by **moments** (or **torques**) that tend to produce **rotation** about the **longitudinal axis** of the bar.

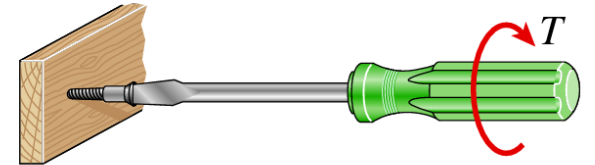


Notations:

- Double-headed arrow
- Curved arrow

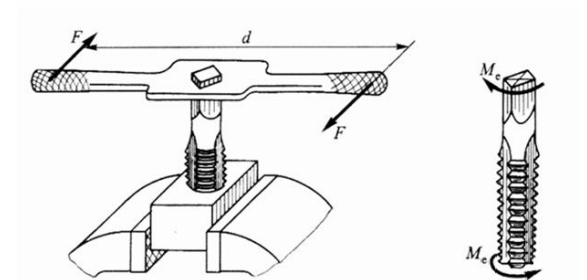


(a)



(b)

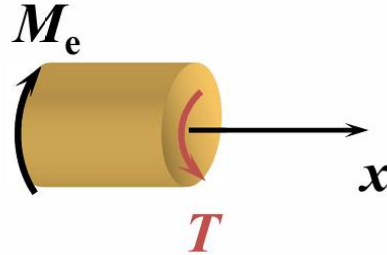
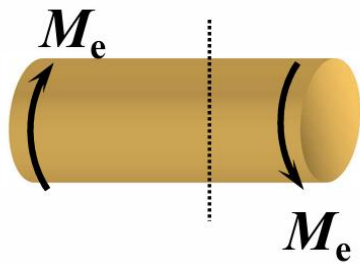
Screwdriver



Screw tap

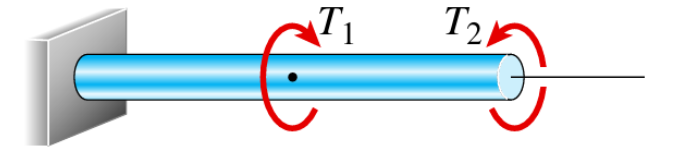
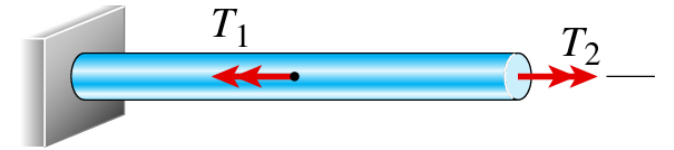
# Torque: the driving force of torsion

Torque / twisting moment / torsional moment



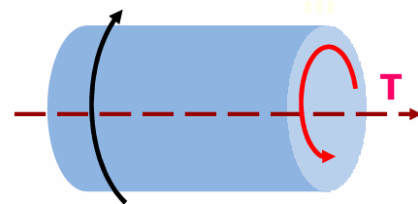
$$T - M_e = 0$$

$$T = M_e$$

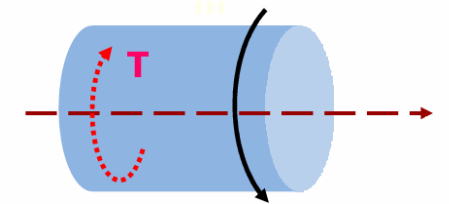


Sign conventions: **right-hand rule**

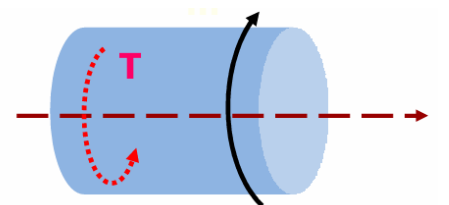
- **Curl** fingers in the direction of twist
- Check where the **thumb** directs
  - outward → positive
  - inward → negative



$T(+)$

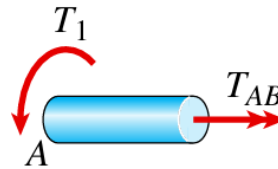
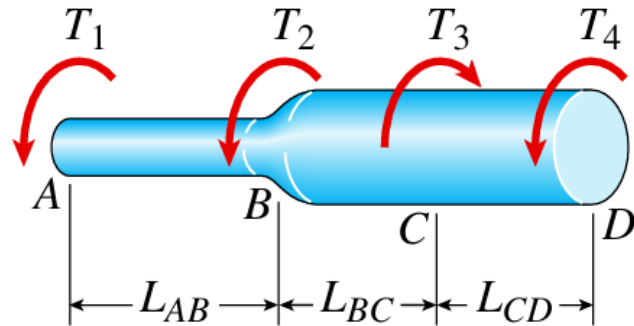


$T(-)$

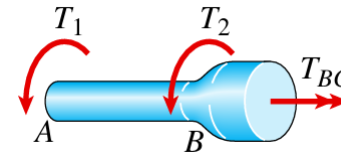


# Torsional moment diagram (TMD)

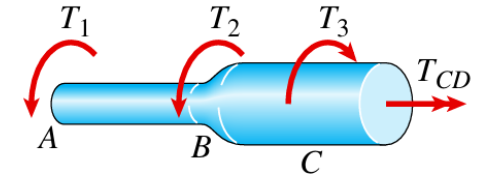
- Always assume the torque in positive direction on each cut
  - Then the sign of magnitude represents the direction
- Apply the method of sections



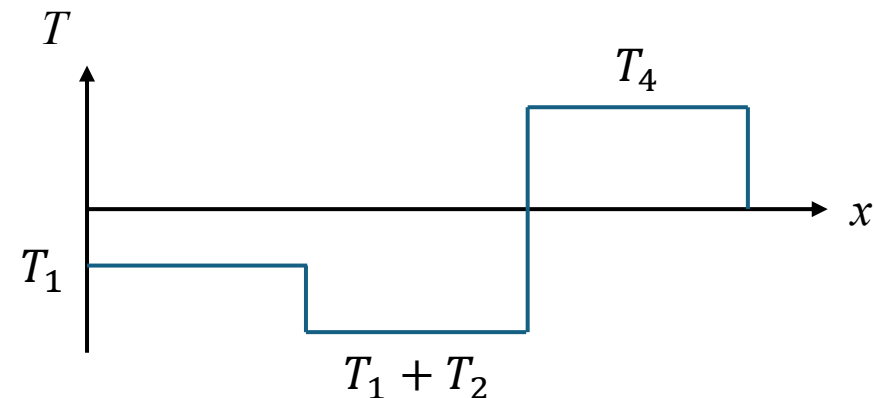
$$T_{AB} + T_1 = 0$$



$$T_{BC} + T_2 + T_1 = 0$$



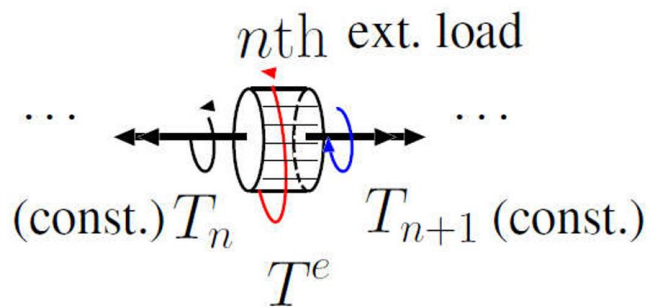
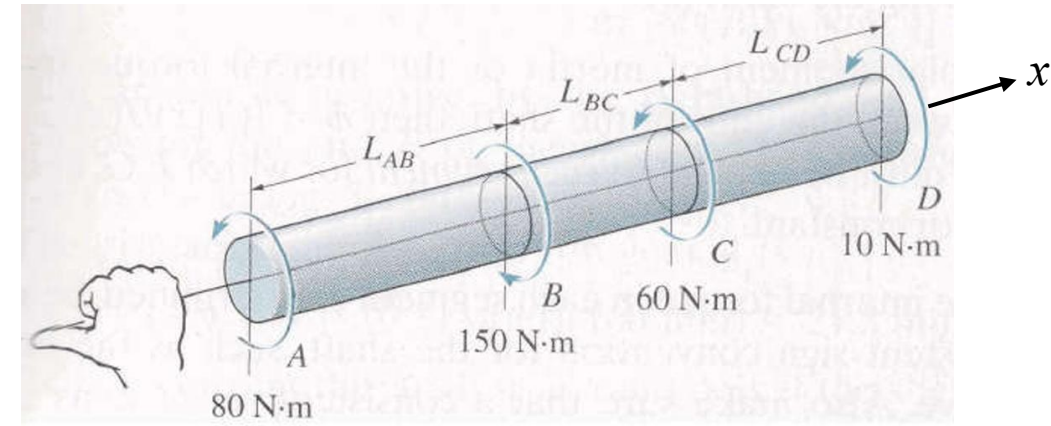
$$T_{CD} - T_3 + T_2 + T_1 = 0$$



# Jump rules for torque

An alternative way for diagram construction:

- ① treating the left- and right-most ends as infinitesimal seg.
- ② number of jumps = number of external loads
- ③ magnitude/direction of jumps = external loads.



sign convention:  
 $T_e$  follows that for  $T_n$

**Equil. Eqn.:**

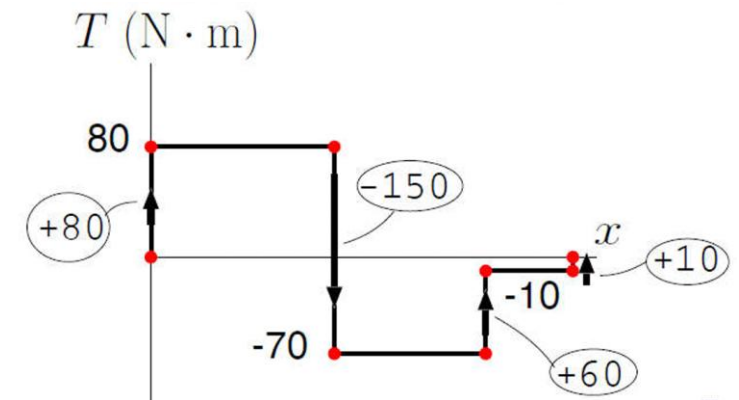
$$T_n + T_e - T_{n+1} = 0 \longrightarrow$$

$$T_{n+1} = T_n + T_e$$

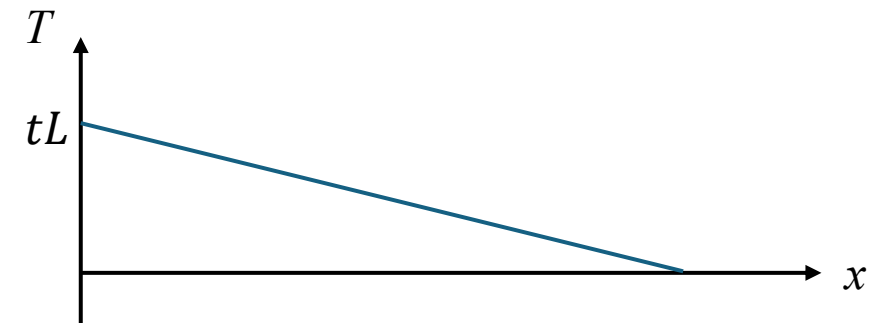
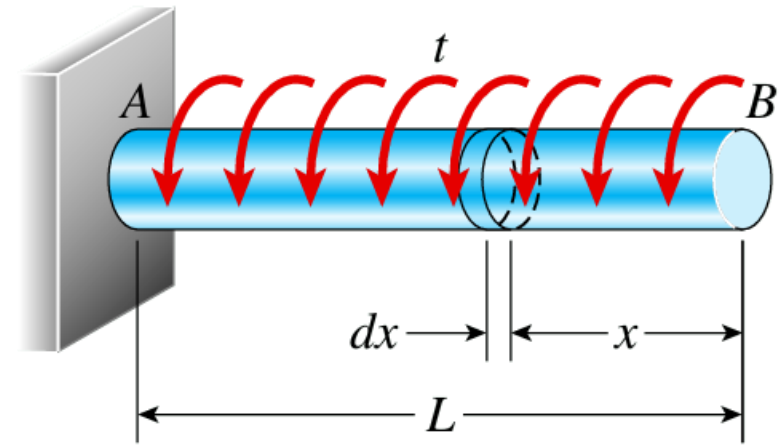
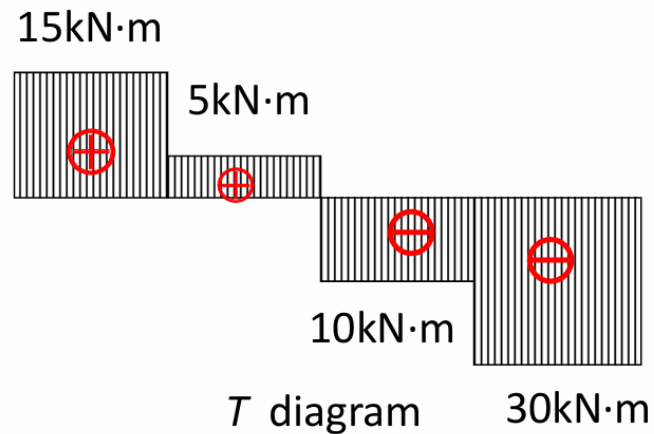
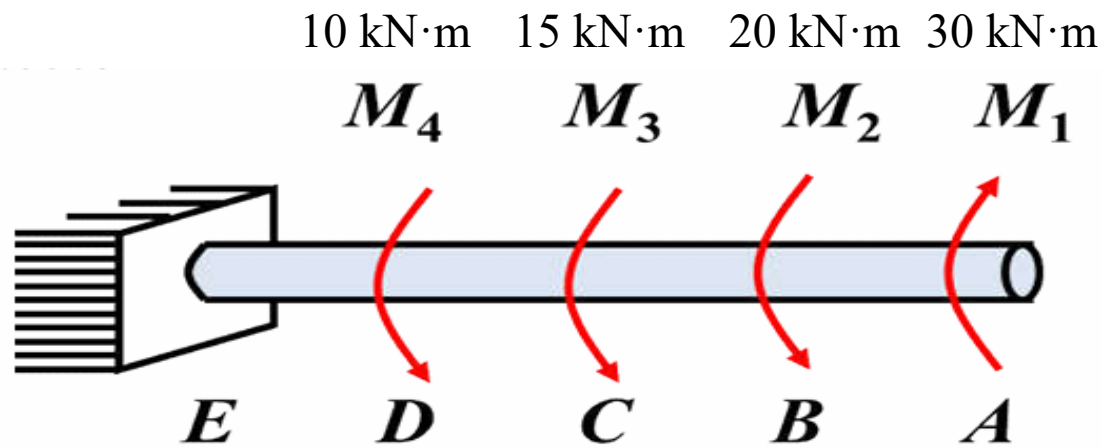
**concentrated torque:**

$$\Delta T = T_{n+1} - T_n = T_e$$

**distributed torque:**  $\frac{dT}{dx} = m_T$

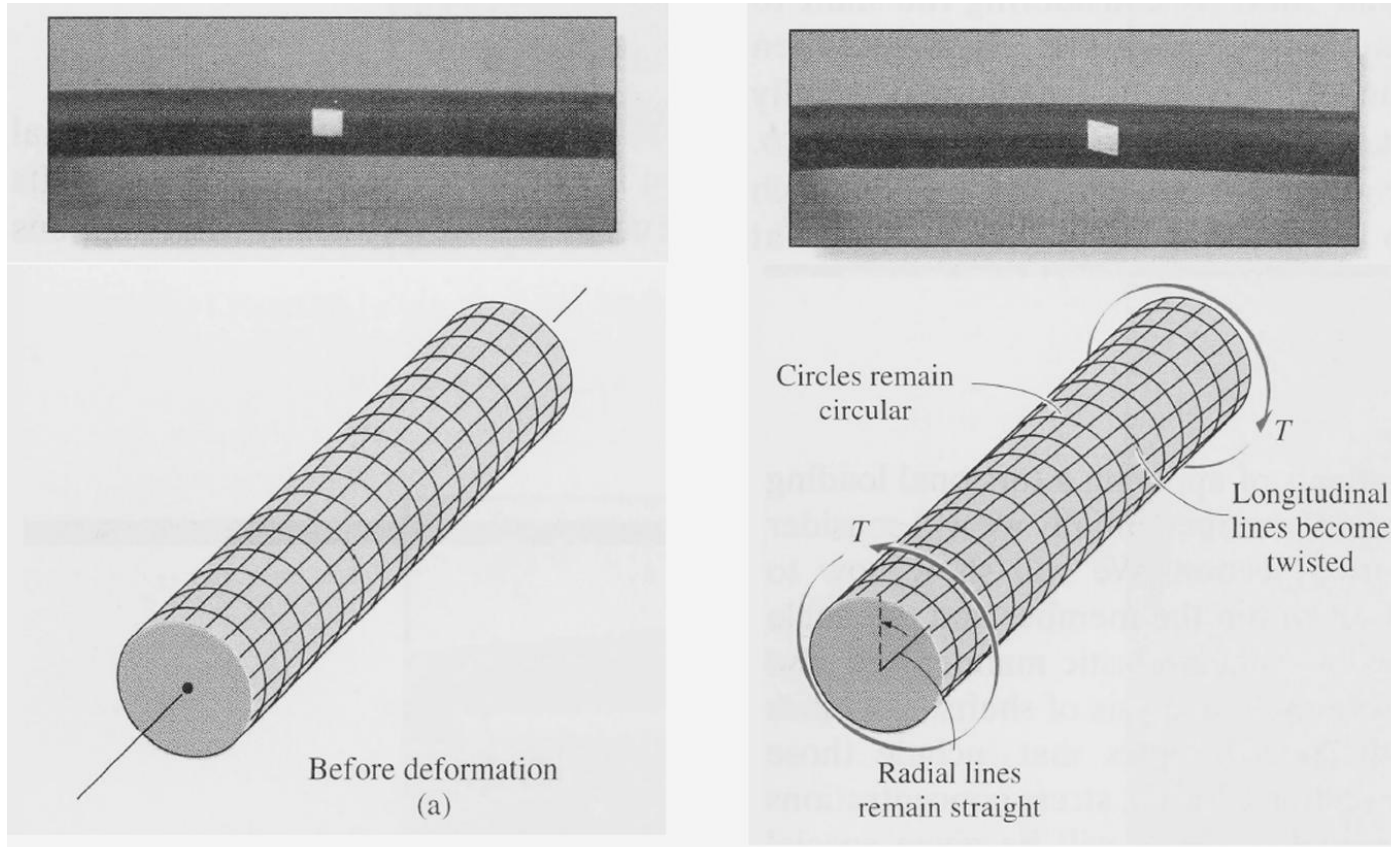


# Examples



# Torsional deformation of a circular shaft

**Pure torsion:** prismatic bars, identical internal torque

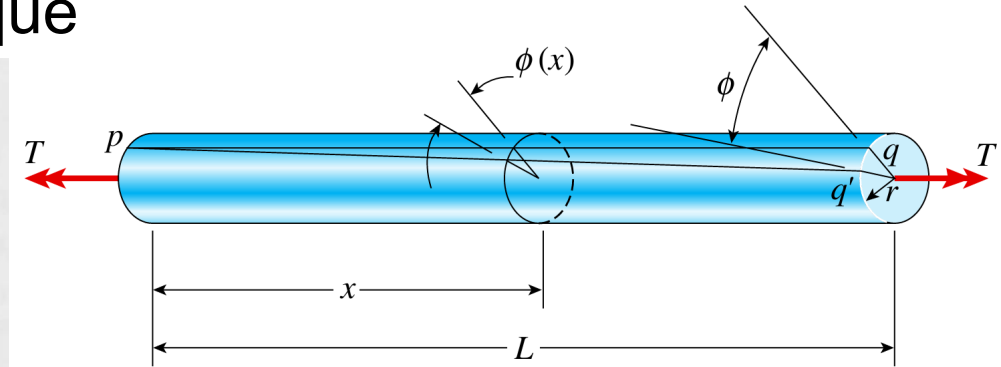


Before deformation

Rectangle

After deformation

Parallelogram



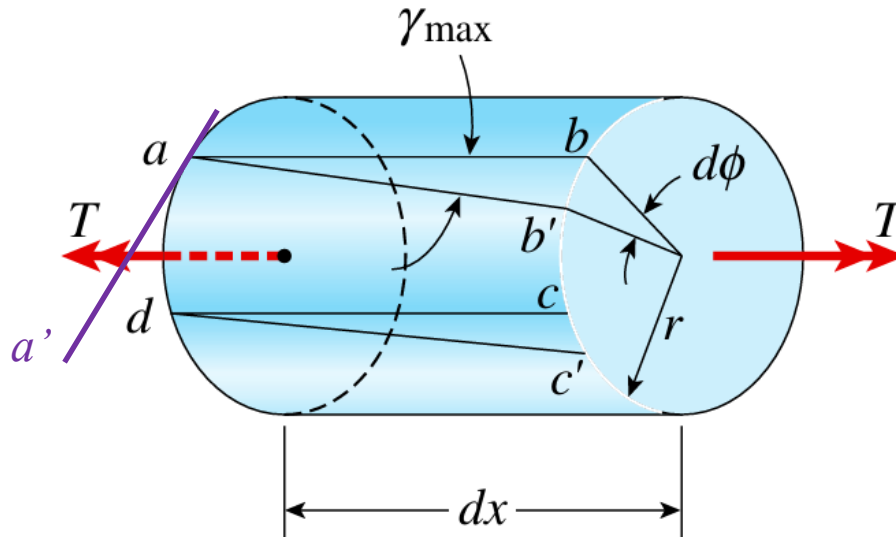
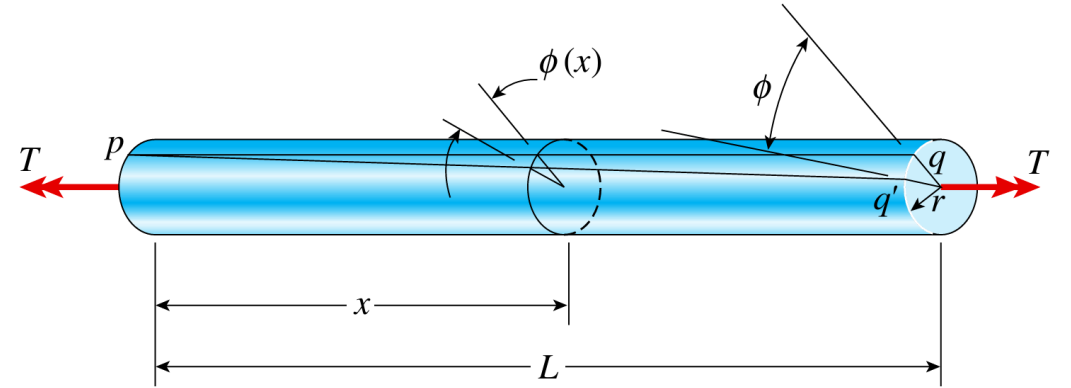
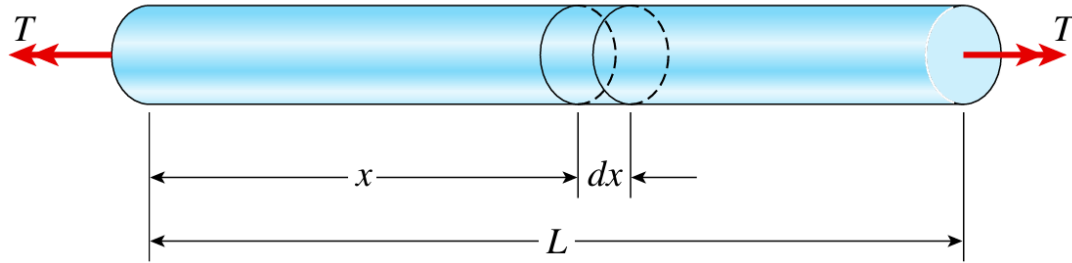
Straight longitudinal line  $\rightarrow$  helical curve

$\phi(x)$ : angle of twist (AoT) – relative rotation

For small angle of twist:

- cross sections remain planar (no warping or bulging)
- radial lines remain straight
- circular circumferences remain circular

# Shear strains



$$\gamma_{\max} = \frac{bb'}{ab} = \frac{rd\phi}{dx}$$

Denote rate of twist:

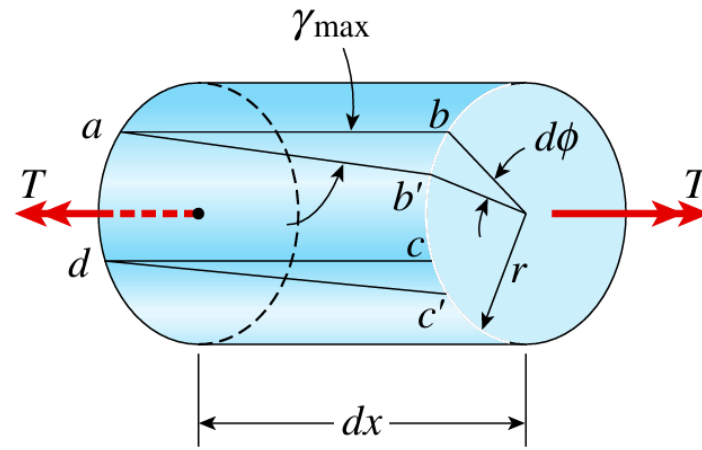
$$\theta(x) = \frac{d\phi(x)}{dx}$$

$$\Rightarrow \gamma_{\max} = r\theta$$

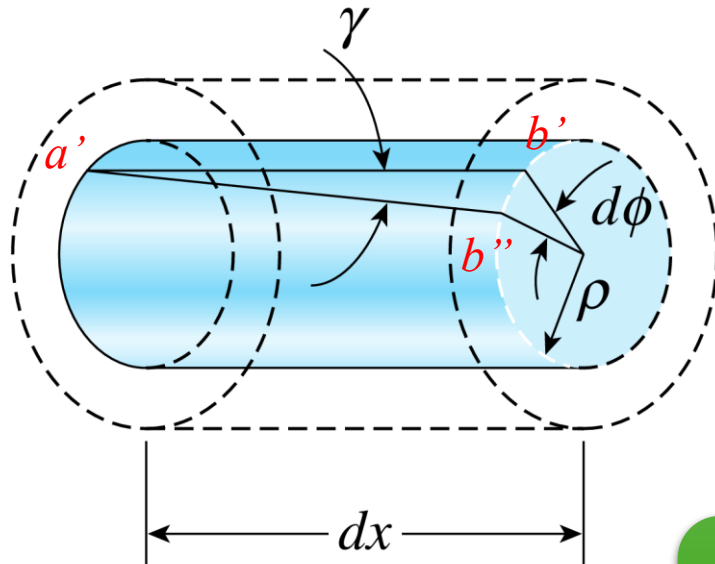
At fixed  $x$ ,

$$\theta = \frac{d\phi}{dx} = \text{const.}$$

# Shear strains



$$\gamma_{\max} = \frac{bb'}{ab} = \frac{rd\phi}{dx} = r\theta$$



$$\gamma = \frac{b'b''}{a'b'} = \frac{\rho d\phi}{dx} = \rho\theta$$

$$\gamma = \frac{\rho}{r} \gamma_{\max}$$

The relation between deformation and strain

Linear distribution of shear strain

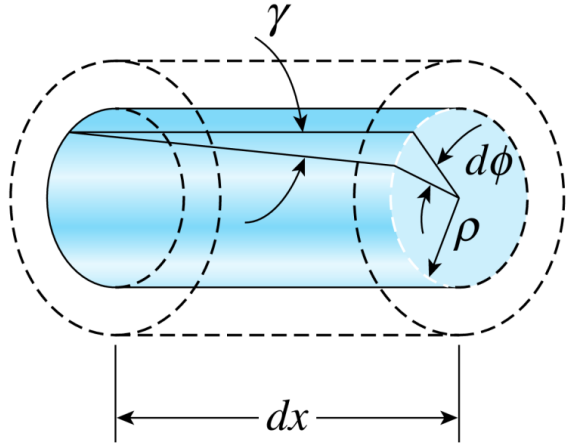
In the special case of **pure torsion**, the rate of twist is a constant

$$\theta(x) = \frac{d\phi(x)}{dx} = \frac{\phi}{L}$$

Therefore, for pure torsion only,

$$\gamma_{\max} = \frac{r\phi}{L}$$

# Circular tubes

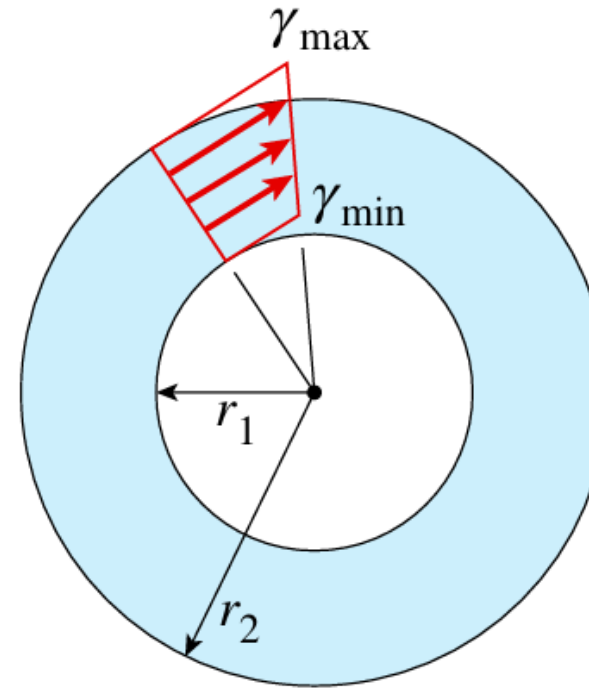


$$\gamma = \frac{\rho d\phi}{dx} = \frac{\rho}{r} \gamma_{\max}$$

For pure torsion,

$$\gamma_{\max} = \frac{r\phi}{L}$$

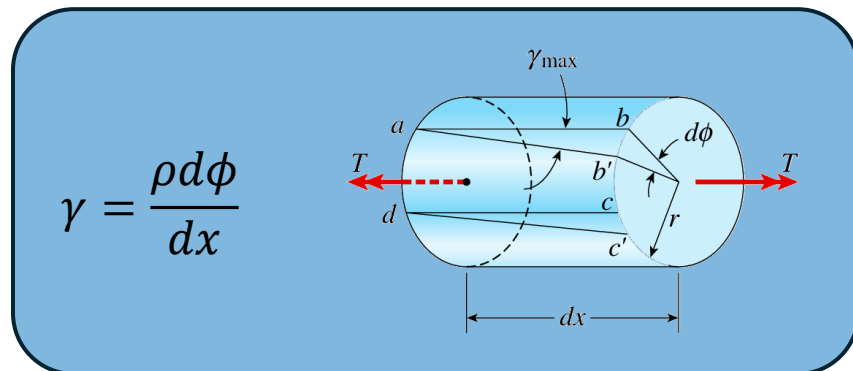
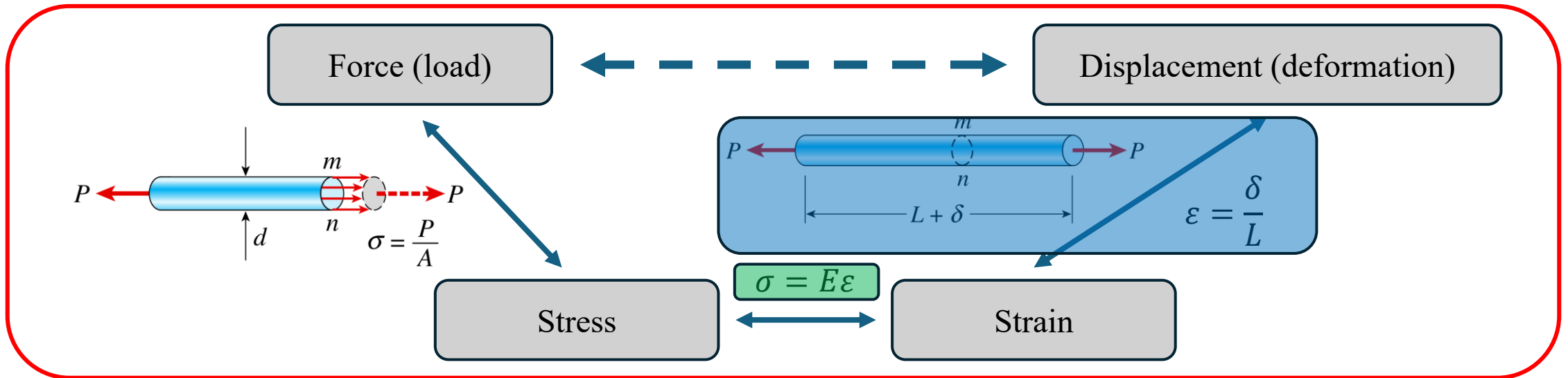
$$\gamma = \frac{\rho\phi}{L}$$



$$\gamma_{\max} = \frac{r_2\phi}{L}$$

$$\gamma_{\min} = \frac{r_1\phi}{L}$$

# Torque-AoT relations



$$\tau = G\gamma$$



$$\tau = G\rho\theta = G\rho \frac{d\phi}{dx} = \frac{\rho}{r} \tau_{\max}$$

$$\tau_{\max} = Gr\theta = Gr \frac{d\phi}{dx}$$